

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('C:\\Users\\751pa\\Downloads\\AirQuality.csv',
delimiter=';')
```

```
df.head()
```

	Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)
PT08.S2(NMHC)	\					
0	10/03/2004	18.00.00	2,6	1360.0	150.0	11,9
						1046.0
1	10/03/2004	19.00.00	2	1292.0	112.0	9,4
						955.0
2	10/03/2004	20.00.00	2,2	1402.0	88.0	9,0
						939.0
3	10/03/2004	21.00.00	2,2	1376.0	80.0	9,2
						948.0
4	10/03/2004	22.00.00	1,6	1272.0	51.0	6,5
						836.0

	N0x(GT)	PT08.S3(N0x)	N02(GT)	PT08.S4(N02)	PT08.S5(O3)	T
RH	\					
0	166.0	1056.0	113.0	1692.0	1268.0	13,6
						48,9
1	103.0	1174.0	92.0	1559.0	972.0	13,3
						47,7
2	131.0	1140.0	114.0	1555.0	1074.0	11,9
						54,0
3	172.0	1092.0	122.0	1584.0	1203.0	11,0
						60,0
4	131.0	1205.0	116.0	1490.0	1110.0	11,2
						59,6

	AH	Unnamed: 15	Unnamed: 16
0	0,7578	NaN	NaN
1	0,7255	NaN	NaN
2	0,7502	NaN	NaN
3	0,7867	NaN	NaN
4	0,7888	NaN	NaN

```
df = df.rename(columns={'T': 'Temperature'})
df = df.rename(columns={'RH': 'Relative Humidity'})
df = df.rename(columns={'AH': 'Absolute Humidity'})
```

```
df.head()
```

	Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)
PT08.S2(NMHC)	\					

0	10/03/2004	18.00.00	2,6	1360.0	150.0	11,9 1046.0
1	10/03/2004	19.00.00	2	1292.0	112.0	9,4 955.0
2	10/03/2004	20.00.00	2,2	1402.0	88.0	9,0 939.0
3	10/03/2004	21.00.00	2,2	1376.0	80.0	9,2 948.0
4	10/03/2004	22.00.00	1,6	1272.0	51.0	6,5 836.0

	N0x(GT)	PT08.S3(N0x)	N02(GT)	PT08.S4(N02)	PT08.S5(03)
Temperature \					
0	166.0	1056.0	113.0	1692.0	1268.0 13,6
1	103.0	1174.0	92.0	1559.0	972.0 13,3
2	131.0	1140.0	114.0	1555.0	1074.0 11,9
3	172.0	1092.0	122.0	1584.0	1203.0 11,0
4	131.0	1205.0	116.0	1490.0	1110.0 11,2

	Relative Humidity	Absolute Humidity	Unnamed: 15	Unnamed: 16
0	48,9	0,7578	NaN	NaN
1	47,7	0,7255	NaN	NaN
2	54,0	0,7502	NaN	NaN
3	60,0	0,7867	NaN	NaN
4	59,6	0,7888	NaN	NaN

```
df = df.drop(['Unnamed: 15', 'Unnamed: 16'], axis=1, errors='ignore')
df['CO(GT)'] = df['CO(GT)'].astype(str).str.replace(',', '.',').astype(float)
df['C6H6(GT)'] = df['C6H6(GT)'].astype(str).str.replace(',', '.',').astype(float)
df['Temperature'] = df['Temperature'].astype(str).str.replace(',', '.',').astype(float)
df['Relative Humidity'] = df['Relative Humidity'].astype(str).str.replace(',', '.',').astype(float)
df['Absolute Humidity'] = df['Absolute Humidity'].astype(str).str.replace(',', '.',').astype(float)

df = df.drop_duplicates()

df = df.fillna(df.mean(numeric_only=True))

df = df.dropna()

df['Absolute Humidity'] = df['Absolute Humidity'].multiply(100)
```

```

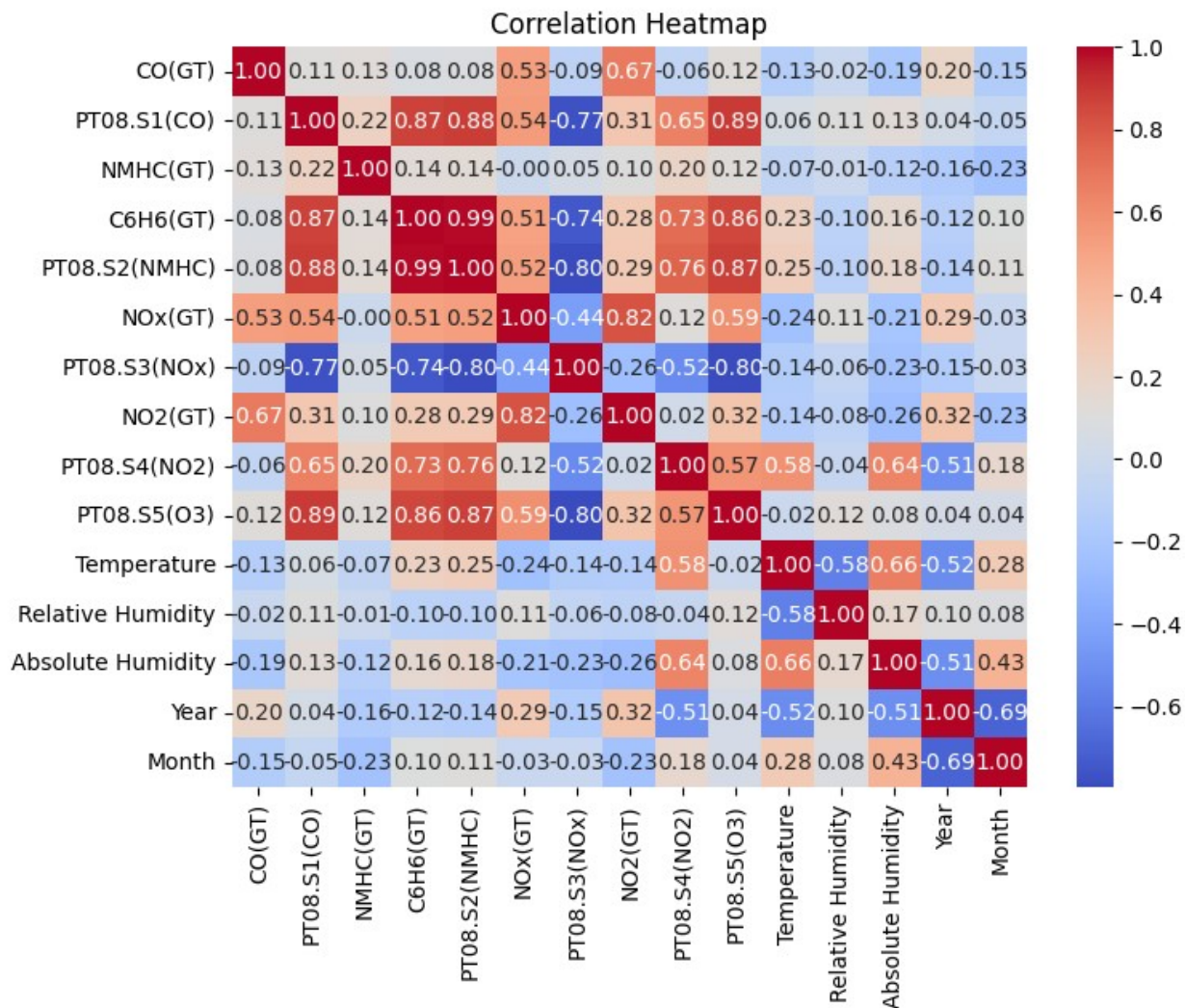
def remove_outliers(column):
    Q1 = column.quantile(0.25)
    Q3 = column.quantile(0.75)
    IQR = Q3 - Q1
    threshold = 1.5 * IQR
    outlier_mask = (column < Q1 - threshold) | (column > Q3 +
threshold)
    return column[~outlier_mask]

cols = ['Temperature', 'Relative Humidity', 'Absolute Humidity',
        'PT08.S4(NO2)', 'PT08.S5(O3)', 'C6H6(GT)', 'PT08.S2(NMHC)',
        'PT08.S1(CO)']
for col in cols:
    df[col] = remove_outliers(df[col])

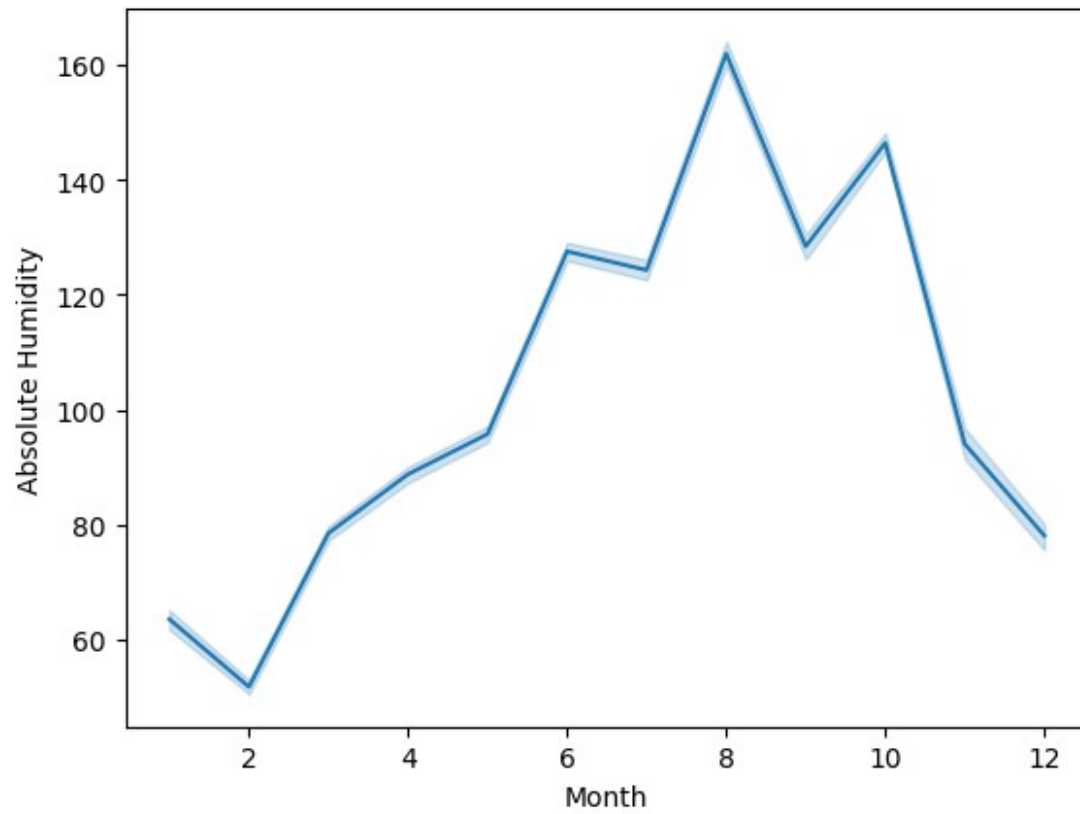
df['Date'] = pd.to_datetime(df['Date'], format='%d/%m/%Y',
errors='coerce')
df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
df['yearr'] = df['Year'].astype(str)
df['month'] = df['Month'].astype(str)

plt.figure(figsize=(8, 6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm',
fmt=".2f")
plt.title('Correlation Heatmap')
plt.show()

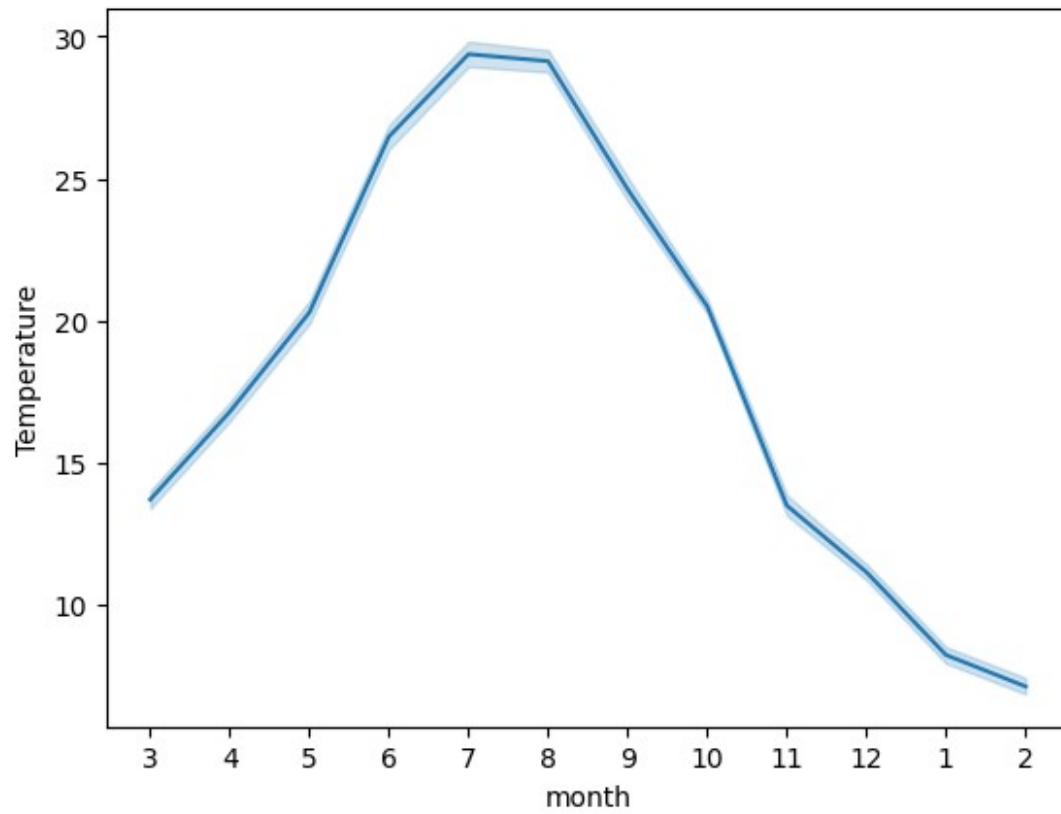
```



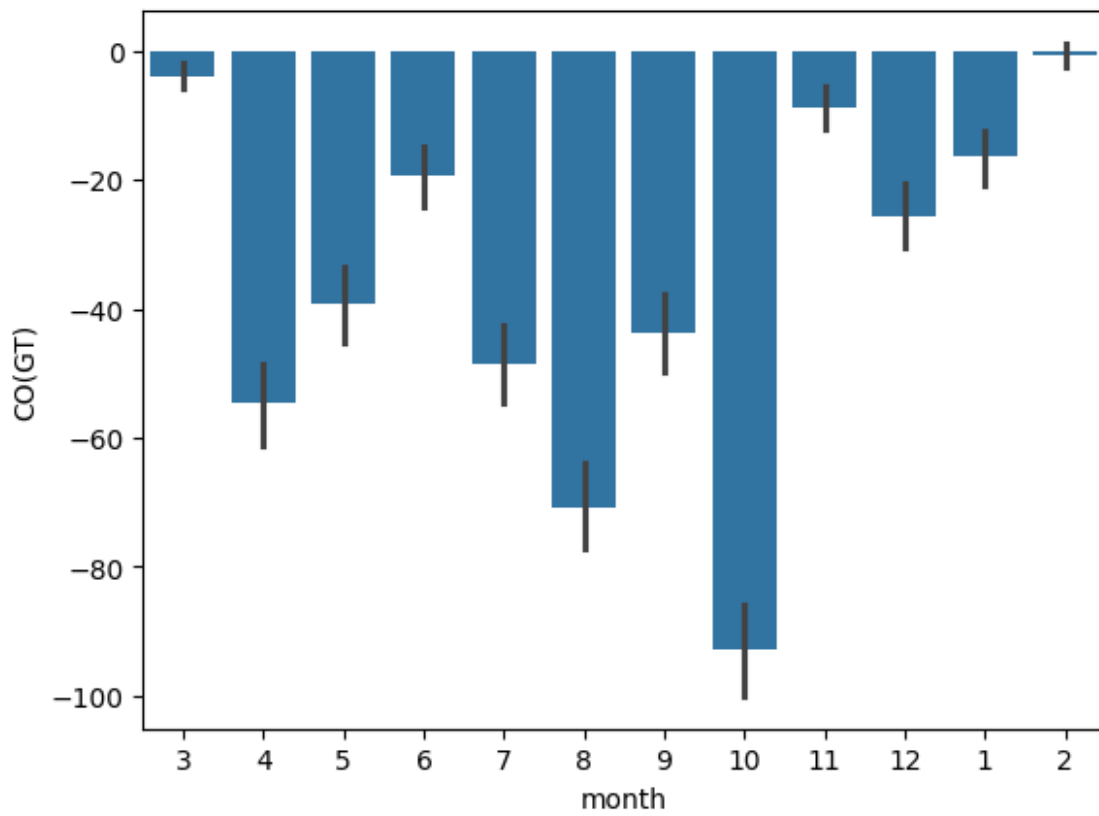
```
sns.lineplot(df, x='Month', y='Absolute Humidity')
plt.show()
```



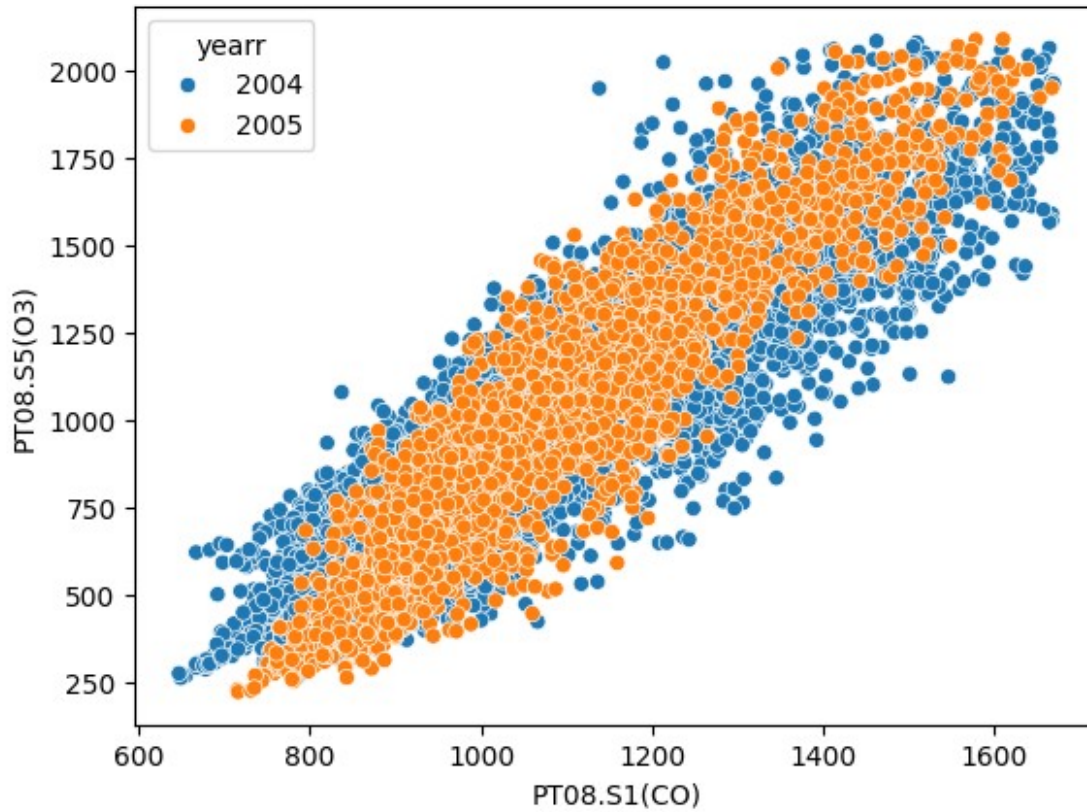
```
sns.lineplot(df, x='month', y='Temperature')  
plt.show()
```



```
sns.barplot(df, x='month', y='CO(GT)')  
plt.show()
```

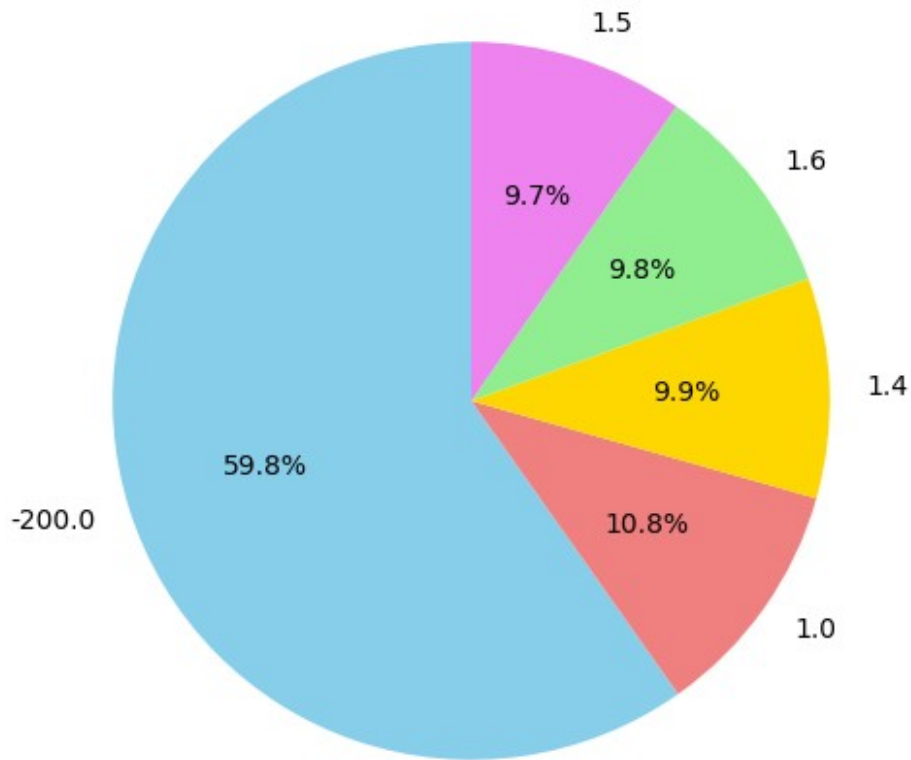


```
sns.scatterplot(df, x='PT08.S1(C0)', y='PT08.S5(03)', hue='yearr')  
plt.show()
```

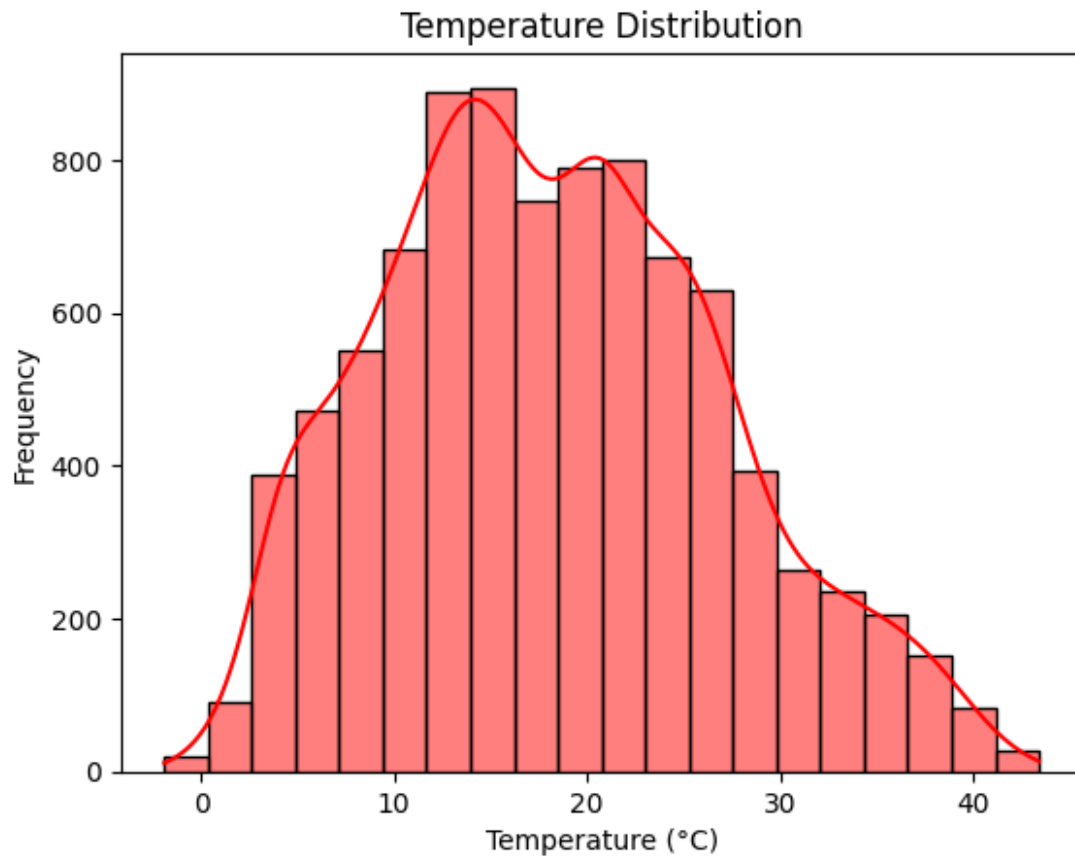


```
co_counts = df['CO(GT)'].value_counts().nlargest(5)
plt.figure(figsize=(6, 6))
co_counts.plot.pie(autopct='%1.1f%%', colors=["skyblue", "lightcoral",
"gold", "lightgreen", "violet"], startangle=90)
plt.title("CO Levels Distribution")
plt.ylabel("")
plt.show()
```

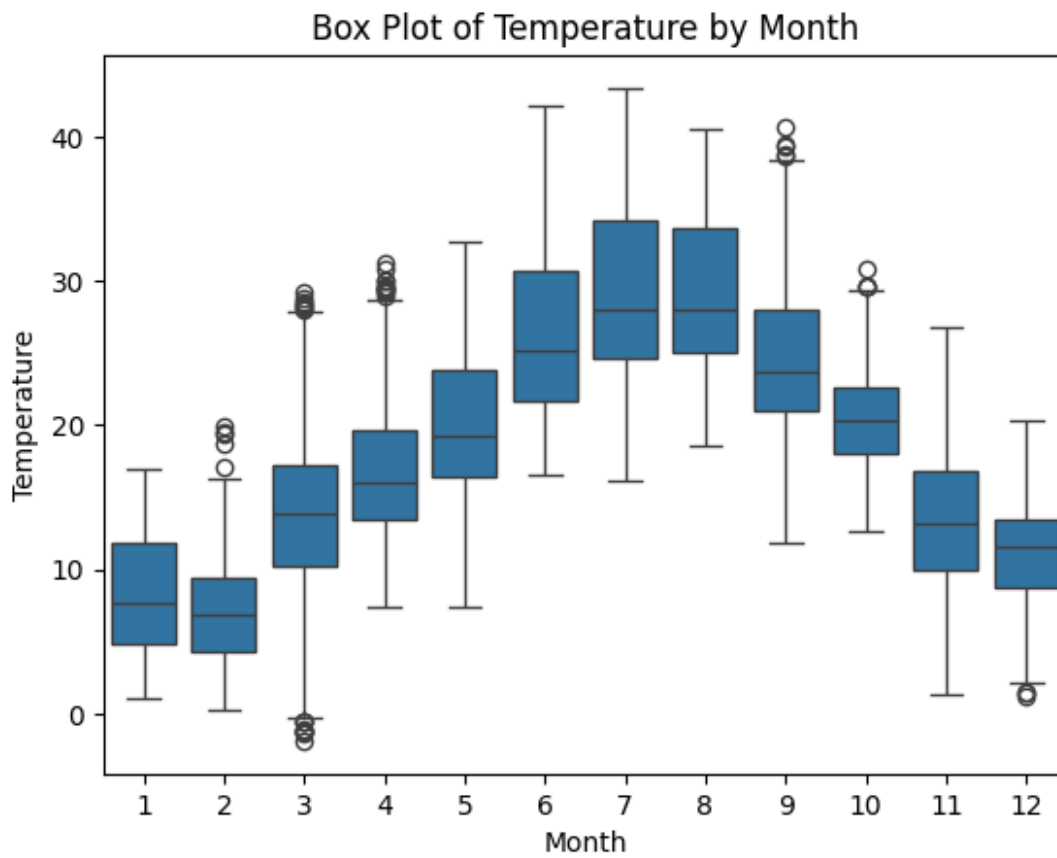
CO Levels Distribution



```
sns.histplot(df['Temperature'], bins=20, kde=True, color='red')  
plt.title("Temperature Distribution")  
plt.xlabel("Temperature (°C)")  
plt.ylabel("Frequency")  
plt.show()
```



```
sns.boxplot(data=df, x='Month', y='Temperature')  
plt.title('Box Plot of Temperature by Month')  
plt.xlabel('Month')  
plt.ylabel('Temperature')  
plt.show()
```



```
df["Category"] = "CO Levels"  
sns.violinplot(x=df["Category"], y=df["CO(GT)"], color="skyblue")  
plt.title("CO Levels Violin Plot")  
plt.xlabel("Category")  
plt.ylabel("CO(GT)")  
plt.show()
```

